

I. Introduction

Public hospitals in the Philippines play an important role in providing accessible and affordable healthcare to a large portion of the population. However, the high demand for government healthcare services can place considerable pressure on hospital facilities, healthcare workers, and existing patient-management processes. One of the resulting concerns is prolonged patient waiting time, particularly when large numbers of patients arrive without appointments and must go through several registration, consultation, diagnostic, and referral processes.

The Philippines National Health Expenditure Survey (NHES) found that long waiting times were the most commonly identified issue during visits to public health facilities, reported by 23.0% of patients. Among patients who sought emergency care in public facilities, the proportion was even higher at 33.8% (Javier et al., 2021). Although this survey predates the most recent health-system developments, it provides nationally representative evidence that waiting time has been a persistent patient-experience problem in the Philippines.

More recent evidence indicates that patient scheduling and service organization remain areas for improvement. A 2025 Philippine Institute for Development Studies (PIDS) baseline assessment found that 68.5% of hospitals accepted walk-in patients for medical consultations without prior appointments, while only 35.5% of hospitals reported being able to schedule medical appointments within two weeks. The report also identified ongoing differences in service readiness and healthcare provider workload across facilities (Conda et al., 2025).

The problem has also been recognized by the Department of Health. In 2025, the DOH began implementing an appointment system in selected public hospitals to address long-standing queues. The system was designed to allow patients to schedule appointments without having to line up extremely early just to get a queueing number.

These developments demonstrate that technology can play a role in improving the organization of patient services. Therefore, this project proposes

eQueue PH, a digital queue and patient-flow management system designed to help public hospitals more efficiently manage patient registration, appointments, queue monitoring, and communication.

II. Problem Description

Long waiting times in public hospitals are influenced by several interconnected factors. A major factor is the large volume of patients seeking services, particularly in hospitals that serve as referral centers for surrounding communities. When patient arrivals exceed the available service capacity, queues can quickly increase, and waiting areas can become crowded.

The problem is also associated with the continued use of walk-in and manual processes. The 2025 PIDS baseline assessment found that more than two-thirds of hospitals surveyed accepted walk-in medical consultations without prior appointments. While walk-in services are important for accessibility, heavy reliance on them can make patient volume difficult to predict and manage. The same assessment found that only 35.5% of hospitals could schedule medical appointments within two weeks, indicating room for improvement in appointment-scheduling systems (Conda et al., 2025).

Waiting time may also increase when patients have to move through multiple hospital services. A 2024 study conducted at the Philippine General Hospital found that waiting times for referrals from family medicine to specialty clinics varied considerably according to specialty, urgency, and purpose of referral. The median waiting time was 11 days for medical specialties and 18 days for surgical specialties, while cardiology referrals had a median waiting time of 125 days (Barbadillo et al., 2024)

Another 2024 study involving the ambulatory care unit of a tertiary government hospital identified prolonged length of stay as an important indicator of patient crowding. The researchers emphasized that understanding the factors associated with prolonged stays is important for staffing and policy decisions aimed at preventing overcrowding (Bontia and Babs-ay, 2024).

The consequences of prolonged waiting extend beyond inconvenience. Patients may experience frustration, uncertainty, and dissatisfaction, while healthcare personnel may face increased workload and difficulty managing patient flow. Long queues can also contribute to overcrowding, making it more difficult for hospitals to provide an organized and patient-centered service.

III. Project Solution

The proposed solution is eQueue PH, a digital queue and patient-flow management system that combines online appointment scheduling, digital registration, real-time queue monitoring, and automated notifications.

The system would allow patients to register via a mobile application, a website, or a hospital kiosk. For non-emergency services, patients could select an available appointment schedule before visiting the hospital. For walk-in patients, the system could generate a digital queue number after registration.

Once registered, patients would be able to monitor their queue status and estimated waiting time through the system. As their turn approaches, the system could send an SMS or an application notification, allowing patients to remain in an appropriate waiting area rather than continuously standing in a physical queue.

The system would also provide hospital personnel with a centralized dashboard. Authorized staff could monitor the number of patients waiting, patients currently being served, queue status, and areas experiencing high patient volume. This information could help hospital administrators identify bottlenecks and make better decisions regarding staffing and patient distribution.

The proposed eQueue PH system would consist of the following major features:

(1) **Digital Registration**

Patients can register through a mobile application, website, or hospital kiosk.

(2) Appointment Scheduling

Patients seeking non-emergency consultations can select available schedules before arriving at the hospital.

(3) Digital Queue Number

The system automatically generates a queue number after registration.

(4) Real-Time Queue Monitoring

Patients can see their queue position and estimated waiting time.

(5) SMS and Application Notifications

Patients receive notifications when their scheduled time or turn in the queue approaches.

(6) Hospital Queue Dashboard

Hospital personnel can monitor patient volume, queues, and service status across departments.

(7) Patient-Flow Monitoring

The system can identify where delays occur between registration, consultation, diagnostics, referrals, and other services.

(8) Data Analytics

The system can generate reports on average waiting times, peak hours, patient volume, and department congestion. These data can support hospital planning and quality-improvement activities.

The system should not replace clinical triage or healthcare professionals' judgment. Emergency patients must continue to undergo appropriate assessment and prioritization by qualified healthcare personnel. The digital system would primarily support administrative processes and patient flow. Therefore, eQueue PH is intended not simply to digitize the queue but to provide hospitals with real-time information to support better patient flow management. By combining appointments, queue monitoring, notifications, and hospital-level analytics, the system can help reduce unnecessary physical waiting and improve the overall patient experience.

IV. Conclusion

Long waiting times remain an important concern in the Philippine public healthcare system. National survey evidence identified lengthy wait times as the most frequently reported issue among patients visiting public health facilities, while more recent Philippine assessments continue to identify challenges in appointment scheduling, service readiness, and healthcare provider workload.

The problem cannot be solved by technology alone because waiting times are also affected by staffing, facility capacity, patient volume, clinical urgency, and the availability of healthcare services. However, information technology can help hospitals organize existing resources more effectively.

eQueue PH aims to transform the patient experience from “waiting without knowing” to “knowing when and where to receive care.” By reducing unnecessary physical queues and providing healthcare personnel with better information on patient demand, the proposed system can contribute to more organized, efficient, and patient-centered public healthcare services.

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